

# A Packing of 95 Lines in $\mathbb{R}^{13}$

Limin Ge\*

October 8, 2026

## Abstract

We give 95 1-dimensional subspaces of  $\mathbb{R}^{13}$  whose smallest squared chordal distance is at least 0.885826. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.885088, obtained by deleting 1 of the 96 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

## 1 Result

**Theorem.** *There are 95 subspaces of dimension 1 in  $\mathbb{R}^{13}$  whose pairwise squared chordal distances  $D(a, b) = 1 - \text{tr}(P_a P_b)$  are all at least*

$$D_{\min} = 0.885826 \dots$$

The exact value of  $D_{\min}$ , a rational number, is given in Appendix A.

## 2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some  $N$  in the row  $m = 13$ ,  $n = 1$ . The cell  $N = 95$  is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.885088, obtained by deleting 1 of the 96 lines of the packing found by this search. The Rankin simplex bound for this cell is  $1(13 - 1)/13 \cdot 95/94 = 0.932897$ , and the value above lies below it.

## 3 Verification

Each subspace is given by an integer  $m \times n$  basis  $Y$  (coordinates times  $10^{12}$ , rounded). With  $G = Y^\top Y$  the projection is  $P = YG^{-1}Y^\top$ , so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n1N95.txt`

in <https://github.com/Ge-limin/math>.

---

\*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

## 4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 82 seconds on one laptop, with about ten such searches running at once.

## A Exact value

numerator:  
80529700758998823207370644153362350398650085706  
denominator:  
90909090909146292876751258212249955615216074405

## B Coordinates

95 blocks of 13 rows with 1 integers each; divide by  $10^{12}$ . File: `grassmann/packings/m13n1N95.txt`.

|               |               |               |               |
|---------------|---------------|---------------|---------------|
| -51124197570  | 350122373792  | 119060773634  | -15598958444  |
| 322721614604  | 32043912436   | -184698037402 | 336508424206  |
| -381054319518 | -83608387099  | 178121987017  | -122379730761 |
| -294835820285 | -40915456654  | -188979691797 | 6014184384    |
| -153220273400 | 336295736261  | -329675105064 | -418239741736 |
| 282677569902  | 176390542187  | -184506373727 | 381132893738  |
| 29911907756   | 399777038892  | -363178112042 | 334112984686  |
| -272938949159 | 119960826105  | -138930275237 | 234018244743  |
| 341673412884  | 251092989589  | 57186565095   | -57680920716  |
| -485322633453 | -553592238825 | 175345699130  | 109909737060  |
| 49028479899   | -223514737887 | 250036464673  | -13908954233  |
| -263406010831 | -281075319773 | 335964417644  | 61719171407   |
| -241387177884 | -113981350303 | -469228230531 | -250725611772 |
| 462718917435  | -235236781779 | 421849215892  | 206759265013  |
| 40248233208   | -119708229234 | 240428514701  | 225205198324  |
| -43873818523  | -102118335975 | 70809122451   | -368791874459 |
| -183912318802 | -96103727409  | -399908289067 | -186797383520 |
| -180673186320 | 610418293480  | -44284909554  | 530373115506  |
| -255443800849 | -81037250950  | -283369544016 | -73398709532  |
| 60155767865   | -135904963884 | 128593112668  | -190187196146 |
| -125307852369 | -39444253824  | 407441177267  | -15978256444  |
| -32567613309  | 41568614684   | 105826240404  | 553519898768  |
| 270844091365  | -289057544880 | -342871141699 | 168862219833  |
| 39589383317   | -431918711341 | -188534378179 | -104650433423 |
| 326058149940  | 48418615880   | -425341074216 | -246204228042 |
| 670081422377  | -686540456908 | 330685032411  | -210499469791 |
| 293406724150  | 2393592379    | -44591649985  | 21309658331   |
| 291459253213  | 34945355106   | 151605398630  | 44951338103   |
| 167533666299  | -432541900238 | 309393196670  | -197579615166 |
| -239677539774 | 186162848763  | 120594787715  | -591926380940 |
| -430519204962 | -46669876974  | 392092473201  | -332550017754 |
| 202233510233  | -108384957784 | 178201084471  | -101115776251 |
| -228171039792 | 125883604255  | 1689829399    | -194899323046 |
| 31436045737   | -41638735815  | 481662782473  | -13995210777  |
| -2269979729   | 392771733120  | -207276863887 | -522177314120 |
| -416792018893 | -312608071889 | -61099100653  | -267051154472 |
| 404211710440  | 333785176580  | -170086168645 | 14254442977   |
| -1471846993   | 179609795309  | 237127530974  | -79840299055  |
| 356434714413  | -136234096081 | -114696554936 | 300682192127  |
| -327497178434 | -244828361093 | -52307404085  | -201120083812 |
| 91183100093   | -53570093283  | 233410534307  | 752183025659  |
| 205182935295  | -260362004336 | -604760663991 | 82675215409   |
| 409393928456  | -405337512565 | 634078338453  | 249436621849  |
| 442127602255  | -195888803566 | 127561461976  | -247266782449 |
| -313374912497 | 100594952194  | 338552064608  | -16278994176  |
| -167418150457 | -386542295217 | 467092371044  | 47542820937   |
| -105857673081 | 304910380269  | 356281112598  | 149217115278  |
| 364931160536  | 429502880274  | -72037762476  | 96612965653   |
| 80137713734   | 166477104258  | 175379611536  | 257585488781  |
| -175390614600 | 160256326590  | 28859629949   | -257520970774 |
| -240768751606 | -252705294126 | 235760631921  | -55754739149  |
| 336874180873  | 164649430275  | 22165913454   | 598131086342  |
| 339462478186  | -128088224080 | -17364112611  | -336452473024 |
| 436752234140  | -383149630429 | 72906491987   | 27351923545   |
| -105134810222 | 47297623765   | 153206140612  | 448594417046  |
| 310205875955  | 175046385604  | 319545970518  | 169028371278  |
| -251514261082 | 259066177008  | 417415245960  | 15930283293   |
| 541971797766  | 187093315513  | 395552793194  | -191903421025 |
| 164629034353  | -518781307066 | 219536277125  | -8248275425   |
| -229062492529 | 322210159213  | -89519737621  | 254607474961  |
| 39422226953   | 316787255364  | 274107783734  | 257439496364  |
| -2814246269   | -561585921741 | -98137765858  | -354800321895 |
| -206053754541 | -69926092790  | -1935002379   | -35250773066  |
| -94439602897  | 90583994689   | -175076378816 | 126087768612  |
| 311813213684  | 90864916733   | -33754683784  | -295172503545 |
| -323563100661 | -232469027216 | 341169670779  | 126954605542  |
| -532301093371 | 399299172402  | -271431199266 | -162592209430 |
| 220941101054  | 483293975132  | -452051228124 | -175119928968 |
| -335221852035 | 186708288224  | -448395766615 | 305902011037  |
| -8726597489   | 31377246822   | -410199381902 | 203788339259  |

|               |               |               |               |
|---------------|---------------|---------------|---------------|
| -517099303932 | 550843435763  | 138552945606  | -26557404916  |
| 289788834903  | 245160237524  | -139679231119 | 64852578764   |
| 20443051630   | -144515297257 | 376796456376  | 320574407871  |
| 438281641631  | -225991929193 | 248484112937  | -77803764727  |
| -300360316075 | -400684588148 | 168666144050  | -334493210    |
| 233568743715  | -139284269502 | 115889650013  | -62106556081  |
| 121994104024  | -154485061992 | 377124389766  | -194139133266 |
| 116283222264  | -73514122825  | -34966845808  | -210926852153 |
| 154643380684  | 370099941350  | -114421399007 | -628735036665 |
| 262236572448  | 224287406048  | -505818779208 | -50542644973  |
| -443129105990 | 42994160216   | -255257598418 | 312775930670  |
| 201519637560  | -27783573176  | -330873878274 | -289730430191 |
| -429026838832 | 248608541993  | 181674928391  | 144670691316  |
| 7400406217    | 458059393161  | 341852496655  | -345601569733 |
| 183772664982  | 206009738912  | -342112143098 | -456608812025 |
| -132785904363 | 370946135712  | 125972755485  | 100793896491  |
| -512705888845 | -13399910284  | -365362296028 | -74758697946  |
| -138037398503 | -258806767036 | -141954826259 | -466091584913 |
| -352837854691 | -417591457370 | -552944106650 | -78066578365  |
| 24124023290   | -30772062324  | 54742938275   | 170167404647  |
| -156493391548 | -340507062865 | 165666531835  | -527225353072 |
| -90136674861  | 123779391736  | 115744922213  | -141978580015 |
| -31442846201  | -250303994829 | -198394131102 | 159233434264  |
| -56517450326  | -15851178207  | -162505047887 | 200673810614  |
| -158765719952 | -368991046397 | -303188094607 | 139095685115  |
| 461679173682  | -192678457617 | 54175945504   | -148094080920 |
| -216155639037 | 203586190953  | -462686455211 | -223045693368 |
| 552805317694  | -315468065961 | 14589587856   | 113789343013  |
| 507676122820  | -346942566271 | 97098299519   | 394622162379  |
| 124276471576  | -472738874773 | 14333284656   | 291262716308  |
| 313621848625  | -58705127187  | -251111121018 | -253288230341 |
| -20268477307  | -421166077193 | 271670319931  | -6605315604   |
| 353059580817  | 157163040037  | -132200393138 | -39078903361  |
| -382231139609 | 241142963122  | -535168649731 | 258710822185  |
| -241952070455 | 252541155538  | 97768402706   | -159030997718 |
| -163307980404 | 261618920371  | -452223700067 | 350174550875  |
| 300157468525  | 455021536842  | -81128581349  | 617168420308  |
| -29538995016  | 98068560905   | -126917513574 | 115512704765  |
| 22813134511   | -303281085803 | 160537274660  | 363775348865  |
| 269282676781  | 216178759586  | 26280170606   | -397368120480 |
| 66719867483   | 238520005131  | -230837569914 | -85111761232  |
| 32218989673   | 43144088145   | 313418817334  | 365871158329  |
| -157650812745 | 11830696751   | -240531795058 | 389413285726  |
| 621690580516  | 137704502653  | -273017042488 | -103218364841 |
| -251385698702 | -167226780179 | 367704920762  | -478639245016 |
| 28716831906   | 609278089435  | -123803228841 | 217196046171  |
| -29714395924  | -188051864210 | 349221778078  | -74193397256  |
| 487641649659  | -95362903316  | 363916088229  | 24224489619   |
| -309940850588 | -229732691268 | -511870225417 | -213803312992 |
| -664414430598 | 403351386039  | -213790717459 | -228511982016 |
| -103454525286 | 435378419029  | 178396830809  | -161471050160 |
| -117343755110 | -118000625447 | -119877756544 | -492062105614 |
| -290011990730 | -204753570938 | 150393198189  | 119841319347  |
| -19696433587  | 124527926426  | -684831275771 | 67035297869   |
| 283294501385  | -652544994188 | 338660198777  | -18007127587  |
| 22806581971   | 78730049788   | 43837398275   | 141761407209  |
| -92838942683  | 7600618922    | 360310381956  | 253585995374  |
| -156926318391 | -233458245885 | 185537673622  | -415758810600 |
| 93224910287   | 138471075924  | 96088129179   | 126265123923  |
| 191341717643  | -93694028379  | 272179132747  | -13204520245  |
| 117543273415  | -225853062131 | -126678204670 | -70327431626  |
| -583600623714 | -457471163788 | -190429538738 | 673387053311  |
| 293896606328  | 44064094220   | 171508110925  | -69177624397  |
| 162918220337  | -255646972984 | -163836437409 | 46244530265   |
| 201259510768  | 143430442261  | -425020522087 | -40091587950  |
| 33025018526   | 209802244896  | 486392294734  | -203919046521 |
| -179752611906 | -32946425631  | -240820759296 | 171187341478  |
| 457412973067  | 556569330254  | 258973558941  | -44530399621  |
| -382676914407 | -286824979004 | 175096333181  | 304919232125  |
| 240409758115  | 66508017068   | 146714201702  | -274978816565 |
| -1283659152   | -423268999233 | -394818446311 | -4757517302   |
| -117749421901 | -171598645990 | -112428665249 | 290205521115  |
| -234397529858 | -40923384863  | -141134971782 | 238156843765  |
| -428747554502 | 39933995455   | -299746527720 | -677113116314 |
| -261796480961 | 366676004693  | -266435891767 | -344983588084 |
| -13571043085  | 7830060354    | 223033823070  | -193413480977 |
| 442459142906  | 234267863470  | 180762464247  | -32670374126  |
| 652517454830  | -573434568422 | 267130494888  | 178120989935  |
| 52895796959   | -301662572778 | -236598193636 | -25487139957  |
| 103547355593  | 121011302065  | 63395305031   | 407710132733  |
| 76800731800   | -221713271707 | -428071872470 | -112518398090 |
| -136400360722 | -503126736788 | 233671784443  | 114393222768  |
| -138057443006 | -79814676440  | 357581128761  | -274525670846 |
| -3798016360   | -152757028924 | -168414833929 | -140516010846 |
| -591833312465 | 153836201930  | -392319766483 | 401702204950  |
| 184386258563  | 137390090542  | 139590491939  | 418399589854  |
| -488136770142 | 53637214114   | 453871169610  | 339105371377  |
| 420134107270  | -215784863163 | 112912709917  | -442107417911 |
| -163738662434 | -149853516092 | -612738218641 | -45380299697  |
| -280420066183 | 438570948773  | 194445668262  | -177783019451 |
| 145961826102  | 70425574164   | 441304406522  | -262880883041 |
| 150890218657  | 254119537654  | 133554184355  | -179585033904 |
| -78586033688  | 241532784592  | -40664288754  | 54958627214   |
| 100470645844  | -98224731905  | -143048861117 | -102048736967 |
| -48212684509  | 658270717729  | 60182370414   | -60770791024  |
| -139191527042 | -256502012044 | 6066337273    | -52758587805  |
| 116006586366  | -247642492859 | -21037952311  | 454467528826  |
| -89369915975  | 23150174256   | 423987498682  | -435165381224 |
| -123570371864 | -193029399200 | 140246753307  | -562975514392 |
| 447003635003  | 162397063748  | 226243094186  | 301720174561  |
| -25133664844  | -448760799891 | -311958482779 | -70936761073  |
| -362756346562 | 274854050661  | 469675615706  | -261473346328 |

|               |               |               |                |
|---------------|---------------|---------------|----------------|
| 38018832861   | 92754011065   | -17387698562  | -141745353033  |
| -612288165211 | -36744561386  | -239146799475 | 120708552421   |
| -57858840657  | -121394476253 | -508406273613 | -19433585376   |
| -105135466052 | 376312655680  | -58492449679  | -349303858977  |
| -73067782633  | -404735234644 | 379052853463  | 659356420584   |
| 365238290934  | 36242065846   | 382101421442  | -118512310811  |
| 83246467196   | -35915657497  | 188500933341  | -191798901077  |
| 136726538728  | -1906028114   | -123954599686 | 156631110540   |
| -575553634716 | -458280223876 | 128503867328  | -48630879585   |
| 269457140561  | -230514234285 | 272359347646  | 23041582774    |
| -30043124081  | -288101156815 | -115054731259 | 76783197933    |
| 171091322494  | 564255130506  | 401512738066  | -19356434193   |
| 66228352903   | -147077676406 | 238499074799  | 25359755640    |
| -88937611065  | 44950330767   | -369239648884 | -504933201567  |
| -244875811933 | -11957485712  | 411866557194  | -312308523975  |
| -162237506653 | 60266566334   | 191028902013  | 268040900925   |
| -469711690410 | -464544971938 | 93525344565   | 455631208689   |
| 113292246822  | -284057042204 | -360368438536 | -293711663485  |
| 149602776028  | 774739419525  | -322424904472 | 497155149012   |
| -180441026401 | 180785360514  | -182537923721 | 16962059166    |
| -522191193694 | 7826437182    | 454471206638  | 125921150372   |
| -192468635018 | 27118731397   | -341338677971 | 183593406261   |
| 94543312028   | -90120509567  | 281984395113  | 330944224571   |
| -360288770756 | 190403897962  | 93796502931   | 7652652857     |
| -130141802199 | 301209965636  | -393838601235 | 45356942264    |
| 276587954339  | -91323258774  | -286296235245 | -793923991093  |
| -288788207307 | 311009015699  | -167808896674 | 293139006036   |
| 91437379052   | -183218784600 | 111712464126  | -151231927648  |
| -391750256651 | 98559223164   | -69828602886  | -480283673     |
| 124933275604  | -135672089930 | -405397099039 | 258654698990   |
| -224060480788 | -98935415192  | -66454595898  | 70555607792    |
| 259720481878  | 99919567603   | 298498861132  | -166731203322  |
| 16137678124   | 209867697387  | -207765658197 | -825599640     |
| -482855860119 | -101142162763 | -4336334212   | -491049270915  |
| 299080824290  | -268532823842 | 119775717543  | -436788974567  |
| -86305283293  | -93709346716  | 107864350210  | 106838234310   |
| -205272475432 | 364095078596  | 310839181981  | -4273056180    |
| 161103825254  | -284190303340 | 463236862530  | -316579882897  |
| 164075249159  | 373944802726  | 113478779791  | -226620635266  |
| 528757717919  | 205650169915  | -495263461814 | 355886797752   |
| 108752001358  | -597129542650 | -148766236187 | -45793117688   |
| -64903389124  | -273195328766 | -132511404009 | 340961012813   |
| -419057301845 | 176356279460  | 327313616928  | -24189196350   |
| -541127693174 | -42603870920  | 354308523527  | -12916552516   |
| 179650293687  | -268737599311 | 134652116822  | -375544464616  |
| 161850567834  | -53962654398  | -175530578214 | -135207873938  |
| -399240295683 | -100912888550 | 230669848788  | -9993467274    |
| -352269089520 | -166972265036 | 255144195742  | -15303156926   |
| 72783000515   | -232888874651 | 357681806857  | 23685573105    |
| 103656229615  | -550666215582 | -105792626974 | -310210593514  |
| 357795178525  | -327677780790 | -195994797087 | -50655807847   |
| 154431995764  | -46607940066  | -347446269240 | -349469833309  |
| 75283332023   | 141673455053  | 124829190457  | 76095758973    |
| -666275707954 | -355475705176 | 507570823277  | -470112942835  |
| -143469297208 | -490345263360 | 429146570246  | -266554254531  |
| 311626598152  | 29412596464   | -99066905889  | 130715933729   |
| 4850008453    | 194619563537  | -259418504325 | 108718252506   |
| 18235527571   | -118934165539 | -77235057751  | 621966387467   |
| -169464163532 | -326856738521 | -178640101613 | -254131334109  |
| 76673062120   | -43015174293  | 494008084460  | 245552478095   |
| 120937506391  | -212027033871 | -92735986858  | -287594663677  |
| -395706705558 | -28646289219  | 378069975168  | -67677368610   |
| -276755466266 | -701013524321 | 46606369491   | -66213271737   |
| -34019054962  | 217982245638  | 684224629157  | 303512967351   |
| 100373389111  | 225183518427  | -59457309418  | 462309083896   |
| 379991852706  | -319121793994 | 123854045993  | -39191048799   |
| 385236510548  | -240172941087 | -135764100436 | 27074407605    |
| 264256299486  | 208531735028  | -51216540166  | -467942360040  |
| -110846001102 | -20190677579  | 181445367893  | 251778664187   |
| -561422005293 | -28733184950  | -152819054433 | 322392201645   |
| -205145987559 | 79483597145   | -237899141010 | -385386863144  |
| -351666103299 | 215712054628  | 19824590719   | 71328676346    |
| -374203970945 | -90596668432  | -209743719753 | -184340707763  |
| -22595512787  | -242865121827 | 214527996408  | 530450213420   |
| 39297318302   | 522782895910  | 193783391159  | -268626111015  |
| -209517291735 | -182560290585 | 143647329108  | -85106730149   |
| -221727977847 | 240726433567  | 131671514316  | -156450228140  |
| 83417093098   | -412983168635 | -423533574477 | -271903988025  |
| -215460806882 | 6462454560    | -174508346250 | 11229464882    |
| 152443298640  | -44161554267  | -419982966357 | 119176202394   |
| -482450912914 | 584465267249  | -561950670499 | -235798635349  |
| 32252427603   | 200231270788  | -246244030086 | -190601925101  |
| -268998088062 | -181847801540 | 121770798973  | 435385769091   |
| -325347725709 | -73887332141  | 173413070811  | 125620402770   |
| 177500787774  | 360075300373  | -63304604236  | -441847334514  |
| -442043952964 | -291443489075 | -66370401101  | -188032748238  |
| 189260215517  | -493670356698 | -538222867770 | -70856232718   |
| 62030101061   | -117015233029 | -399569368768 | -82956709382   |
| -181813830106 | 54866063047   | 339776853765  | -276072035264  |
| 193492611293  | 57617094946   | -100439800049 | -381068223661  |
| -428049812221 | 193696770259  | 177851849738  | -386372440966  |
| -210836368837 | -11394224055  | 479530867058  | 440274766401   |
| -63088539141  | 220314663926  | 209374357044  | 421430809723   |
| -803583691934 | -597382732861 | -176503711675 | -175309100878  |
| -33167719663  | 201439042466  | 110404055827  | 333991644560   |
| 106626624042  | 346913088931  | 194622799521  | -144163964426  |
| 246497430538  | 583231017302  | 414137510906  | 192610546333   |
| 142812075927  | -126267109670 | -92021206248  | -1020605044803 |
| 76683277068   | -57840196438  | -84504387408  | -15983012076   |
| -243332855959 | -76716839823  | -62338595477  | 141392229151   |
| -122334263670 | -51591005580  | 363295184404  | 415835386902   |
| -389744166535 | -89478094968  | -185406880046 | 607907077718   |

-174925224306  
509810497446  
-483257355196  
205276762277  
-126736036631  
336796066278  
-59303624876  
-21795105968  
321668842167  
6353063337  
-157996035869  
-575838103188  
-100920500262  
-196432189027  
-190975069226  
-51594784367  
390323167271  
348822671256  
-24825015151  
-371105944948  
-265551226103  
-269481967346  
-156490968079  
55354944812  
-450039341280  
212115887490  
-35452227994  
238756101958  
-92235219704  
-40207359221  
528654975697  
93786048257  
285010193672  
67314514867  
531505958389

-171850835511  
-166796559931  
487381313444  
210170251753  
354452487332  
286256308150  
-295001819432  
29926558856  
-81099423137  
-210217893078  
-379883277257  
390272419089  
-134315531284  
192328204885  
-57651984326  
-70168698957  
111023230800  
315191491654  
494857444574  
166835617729  
-543759556441  
-214566040540  
137934455472  
-147809598889  
229342108960  
-367679453557  
79828304243  
-50445858798  
-99222338855  
572343396012  
-108249707827  
-44499787585  
-297414531392  
115156858967  
-34710098057

-240489328331  
-392592959538  
248759508036  
513025782032  
-191098983565  
32045347882  
412336393902  
-220990926293  
-77586901071  
-315671917379  
7831639140  
-391066689154  
-19243921166  
-587241685762  
314961172666  
-82369024058  
-183642816173  
-507750617799  
474355477941  
83415759118  
-367316704685  
191901866536  
-290786972825  
183259763199  
3508229043  
-56372849959  
-102793029686  
-328257704212  
-272760211997  
-156272336166  
-213140621854  
23452661442  
-82951111115  
-470743722656  
469936074300

202368173556  
50913345552  
3395163715  
-356363928919  
-428404676393  
71709848932  
270578838686  
-268817771499  
272420326147  
-182344755314  
-464066658346  
445578600154  
212349288685  
-147306808314  
233598908453  
-453627103490  
-64396024762  
231586370166  
19742506444  
-128565315185  
277135322352  
254005519325  
-147902849530  
128903735243  
52125765490  
340871580132  
-52969555280  
290761186225  
-240398525121  
174778768061  
1112593900  
-167829654571  
-725186829459  
-219837127069